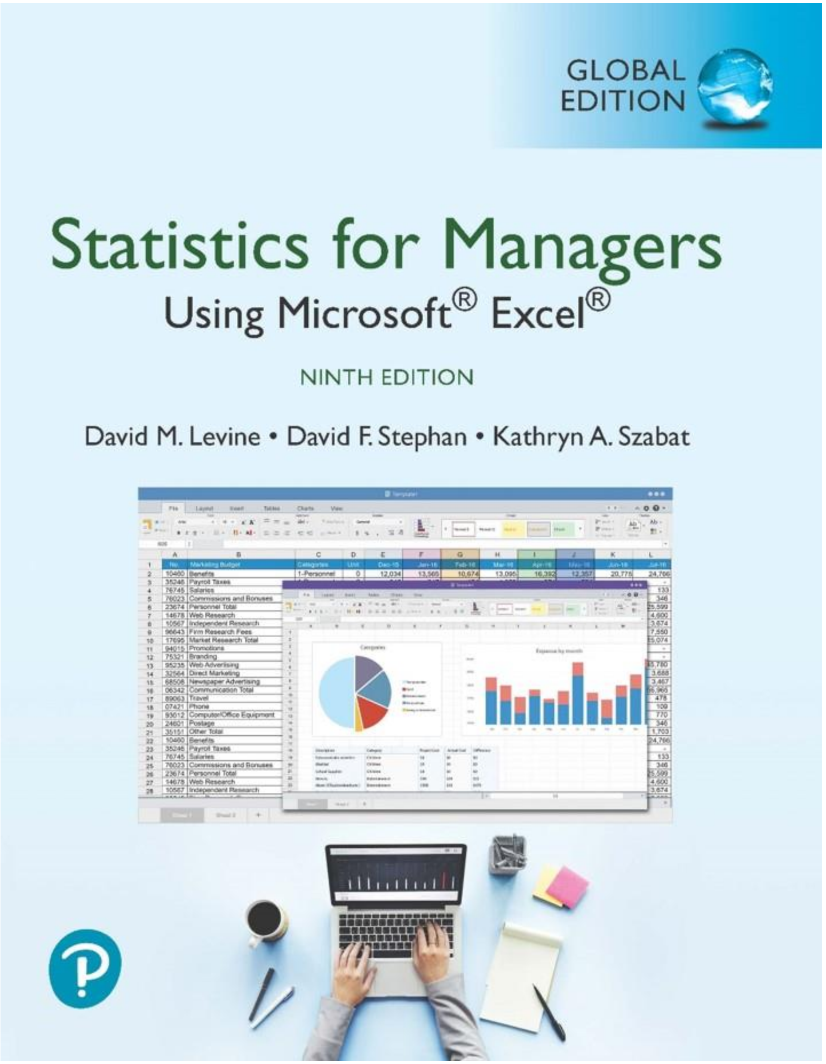

Chapter 3

Numerical Descriptive Measures

Objectives

In this chapter, you learn to:

- Learn to use the sigma (Σ) notation.
- Describe the properties of central tendency, variation, and shape in numerical variables.
- Construct and interpret a boxplot.
- Compute descriptive summary measures for a population.
- Calculate the covariance and the coefficient of correlation.

Sigma (Σ) Notation

- The symbol Σ is pronounced "SIGMA". It is a Greek letter that is often use in mathematical contexts to denote "summation".

- Consider a variable X that contains a set of n values.

- That is, $X = \{x_1, x_2, x_3, \ldots, x_n\}$. Then,

$$\sum X = \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_i + \cdots + x_n$$

- If $X = \{3, 11, 0, 6, 4\}$ then

$$\sum_{i=1}^{5} x_i = 3 + 11 + 0 + 6 + 4 = 24$$

Popular Σ Notation Applications

$$\sum X^2 = \sum_{i=1}^{n} x_i^2 = x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2 \neq \left(\sum_{i=1}^{n} x_i\right)^2$$

Example:

						Total (Σ)
X	3	11	0	6	4	**24**
$\mathbf{X^2}$	9	121	0	36	16	**182**

Therefore,

$$\sum X^2 = 182 \neq 576 = 24^2 = \left(\sum X\right)^2$$

Popular Σ Notation Applications

$$\sum \mathrm{XY} = \sum_{i=1}^{n} x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3 + \cdots + x_n y_n \neq \left(\sum \mathrm{X}\right)\left(\sum \mathrm{Y}\right)$$

Example:

						Total (Σ)
X	3	11	0	6	4	**24**
Y	10	1	2	8	5	**26**
XY	30	11	0	48	20	**109**

Therefore,

$$\sum \mathrm{XY} = 109 \neq 624 = 24 \times 26 = \left(\sum \mathrm{X}\right)\left(\sum \mathrm{Y}\right)$$

Some Σ Notation Rules

$$\sum(X - Y) = \sum X - \sum Y$$

Example:

						Total (Σ)
X	8	11	0	6	4	**29**
Y	10	1	2	8	5	**26**
X – Y	-2	10	-2	-2	-1	**3**

Therefore,

$$\sum(X - Y) = 3 = 29 - 26 = \sum X - \sum Y$$

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Some Σ Notation Rules

$$\sum(X+Y) = \sum X + \sum Y$$

Example:

						Total (Σ)
X	8	11	0	6	4	**29**
Y	10	1	2	8	5	**26**
X + Y	18	12	2	14	9	**55**

Therefore,

$$\sum(X+Y) = 55 = 29 + 26 = \sum X + \sum Y$$

Some Σ Notation Rules

$$\sum cX = c\sum X \qquad \text{(where } c \text{ is a constant)}$$

Example:

						Total (Σ)
X	8	11	0	6	4	**29**
3 × X	24	33	0	18	12	**87**

Therefore,

$$\sum 3X = 87 = 3 \times 29 = 3\sum X$$

Some Σ Notation Rules

$$\sum c = \sum_{i=1}^{n} c = nc \qquad \text{(where } c \text{ is a constant)}$$

Example:

						Total (Σ)
c	8	8	8	8	8	**40**

Therefore,

$$\sum_{i=1}^{5} 8 = 8 + 8 + 8 + 8 + 8 = 40 = 5 \times 8$$

Summary Definitions

DCOVA

- The **central tendency** is the extent to which the values of a numerical variable group around a typical or central value.

- The **variation** is the amount of dispersion or scattering away from a central value that the values of a numerical variable show.

- The **shape** is the pattern of the distribution of values from the lowest value to the highest value.

Measures of Central Tendency: The Mean

DCOVA

- The arithmetic mean (often just called the "mean") is the most common measure of central tendency.

  - For a sample of size n:

Pronounced x-bar

The i^{th} value

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

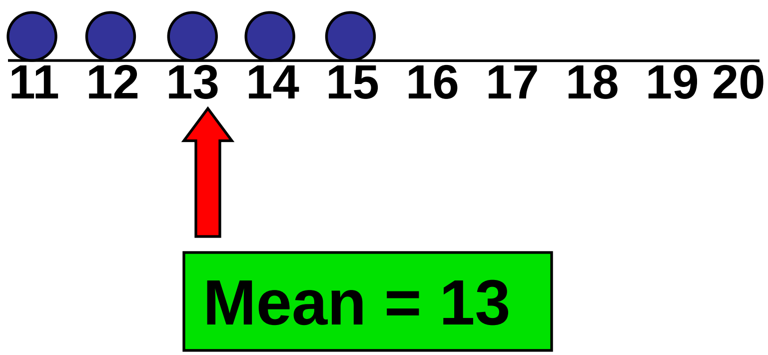
Sample size

Observed values

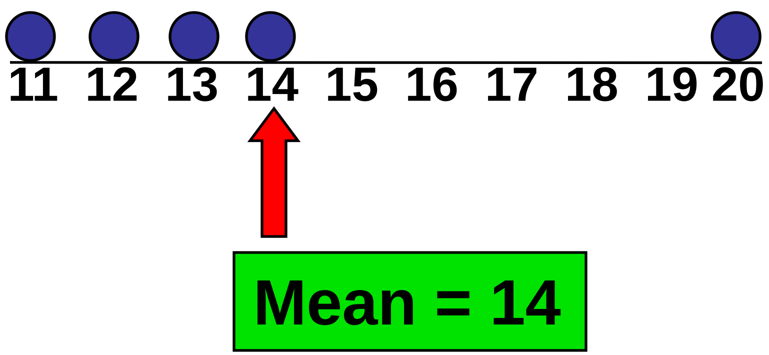
Measures of Central Tendency: The Mean (cont'd)

DCOVA

- The most common measure of central tendency.
- Mean = sum of values divided by the number of values.
- Affected by extreme values (outliers).



$$\frac{11+12+13+14+15}{5} = \frac{65}{5} = 13$$

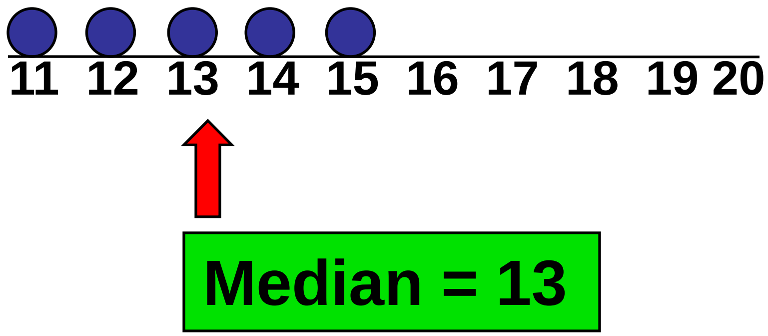


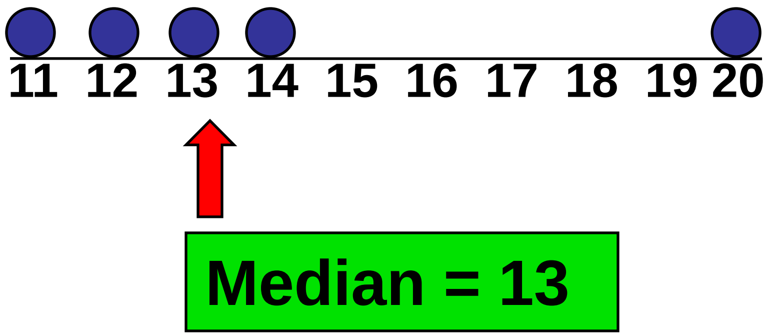
$$\frac{11+12+13+14+20}{5} = \frac{70}{5} = 14$$

Measures of Central Tendency: The Median

DCOVA

- In an ordered array, the median is the “middle” number (50% above, 50% below).





- Less sensitive than the mean to extreme values.

Measures of Central Tendency: Locating the Median

DCOVA

- The location of the median when the values are in numerical order (smallest to largest):

$$\text{Median position} = \frac{n+1}{2} \text{ position in the ordered data}$$

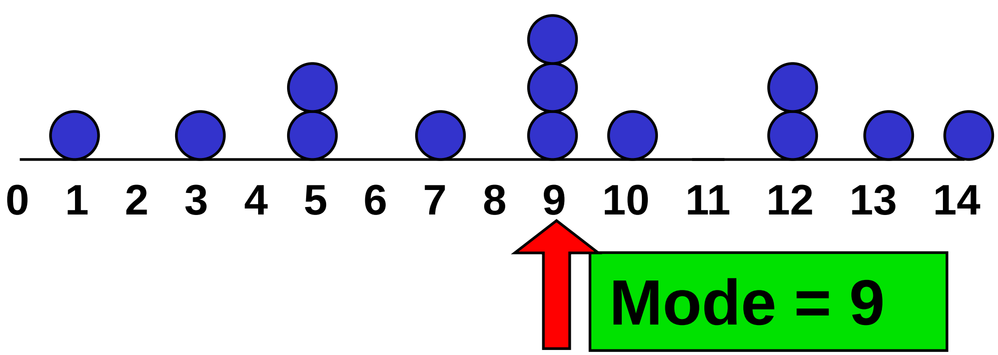
- If the number of values is odd, the median is the middle number.
- If the number of values is even, the median is the average of the two middle numbers.

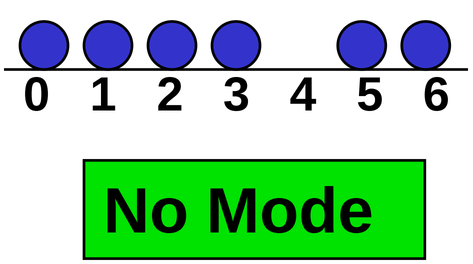
Note that $\frac{n+1}{2}$ is not the *value* of the median, only the *position* of the median in the ranked data.

Measures of Central Tendency: The Mode

DCOVA

- Value that occurs most often.
- Not affected by extreme values.
- Used for either numerical or categorical data.
- There may be no mode.
- There may be several modes.





Measures of Central Tendency: Review Example

DCOVA

House Prices:
R2,000,000
R 500,000
R 300,000
R 100,000
R 100,000
Sum R **3,000,000**

- **Mean:** (R3,000,000/5)
  = **R600,000**
- **Median:** middle value of ranked data
  = **R300,000**
- **Mode:** most frequent value
  = **R100,000**

Measures of Central Tendency: Which Measure to Choose?

DCOVA

- The **mean** is generally used, unless extreme values (outliers) exist.

- The **median** is often used, since the median is not sensitive to extreme values. For example, median home prices may be reported for a region; it is less sensitive to outliers.

- In many situations it makes sense to report both the **mean** and the **median**.

Measuring Central Tendency For The Rate Of Change Of A Variable Over Time: Geometric Mean & Geometric Mean Rate of Return

DCOVA

- Geometric mean
  - Used to measure the rate of change of a variable over time.

$$\bar{x}_G = (x_1 \times x_2 \times \cdots \times x_n)^{1/n}$$

- Geometric mean rate of return
  - Measures the status of an investment over time.

$$\bar{r}_G = [(1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_n)]^{1/n} - 1$$

  - Where r_i is the rate of return in time period i.

The Geometric Mean & The Geometric Mean Rate of Return: Example

DCOVA

An investment of R100,000 declined to R50,000 at the end of year one and rebounded to R100,000 at end of year two:

$$x_1 = \text{R}100,\underbrace{000 \quad x_2 = \text{R}50,}_{50\%\ \text{decrease}}\underbrace{000 \quad x_3 = \text{R}100,}_{100\%\ \text{increase}}000$$

The overall two-year return is zero, since it started and ended at the same level.

The Geometric Mean & The Geometric Mean Rate of Return: Example

$$\text{Rate} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}}$$

$$\text{Rate}_{y_1} = \text{Not Applicable}$$

$$\text{Rate}_{y_2} = \frac{\text{Value}_{y_2} - \text{Value}_{y_1}}{\text{Value}_{y_1}} = \frac{\text{R50,000} - \text{R100,000}}{\text{R100,000}} = -0.50$$

$$\text{Rate}_{y_3} = \frac{\text{Value}_{y_3} - \text{Value}_{y_2}}{\text{Value}_{y_2}} = \frac{\text{R100,000} - \text{R50,000}}{\text{R50,000}} = +1.00$$

	Year 1	**Year 2**	**Year 3**
Value	R100,000	R50,000	R100,000
Rate	-	-0.50	+1.00

The Geometric Mean & Geometric Mean Rate of Return: Example (cont'd)

- Use the 1-year returns to compute the arithmetic mean and the geometric mean:

Arithmetic mean rate of return:

$$\bar{x} = \frac{-0.50 + 1.00}{2} = 0.25 = 25\%$$

Misleading result

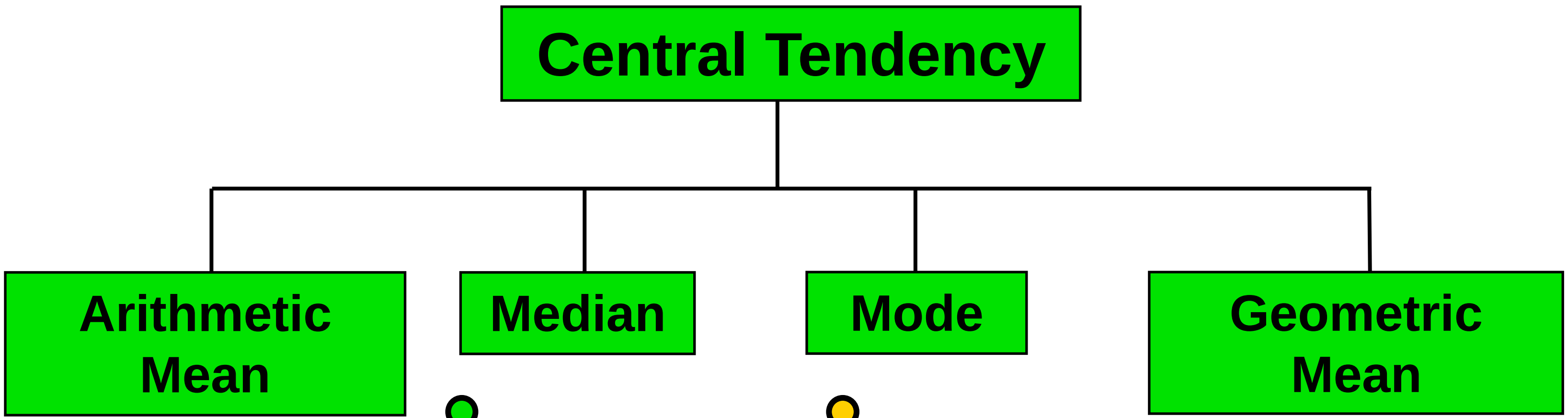
Geometric mean rate of return:

$$\bar{r}_G = [(1 + (-0.50)) \times (1 + 1.00)]^{1/2} - 1$$
$$= (0.50 \times 2.00)^{(1/2)} - 1 = 1^{(1/2)} - 1 = 0\%$$

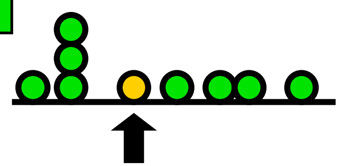
More representative result

Measures of Central Tendency: Summary

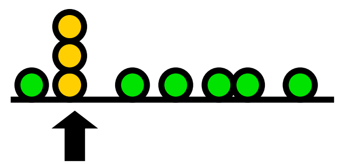
DCOVA



$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$



Middle value in the ordered array



Most frequently observed value

$$\bar{x}_G = (x_1 \times x_2 \times \cdots \times x_n)^{1/n}$$

Rate of change of a variable over time

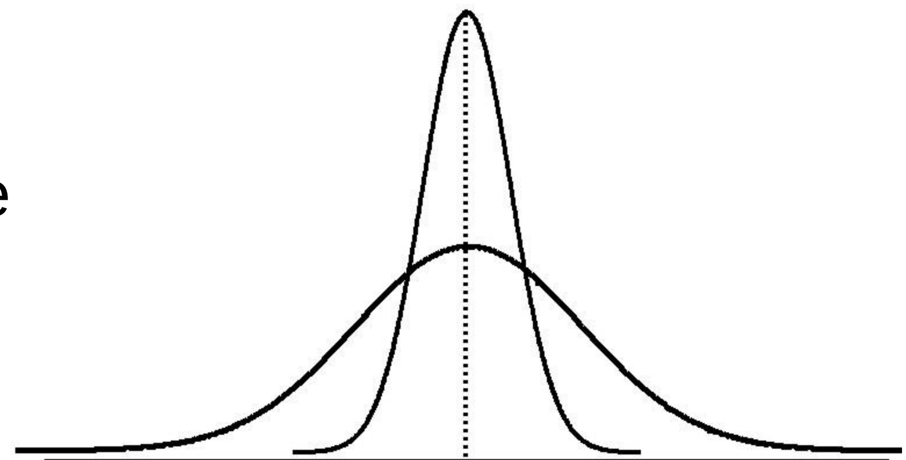
Measures of Variation

DCOVA

Variation

- **Range**
- **Variance**
- **Standard Deviation**
- **Coefficient of Variation**

- Measures of variation give information on the **spread** or **variability** or **dispersion** of the data values.



Same center, different variation

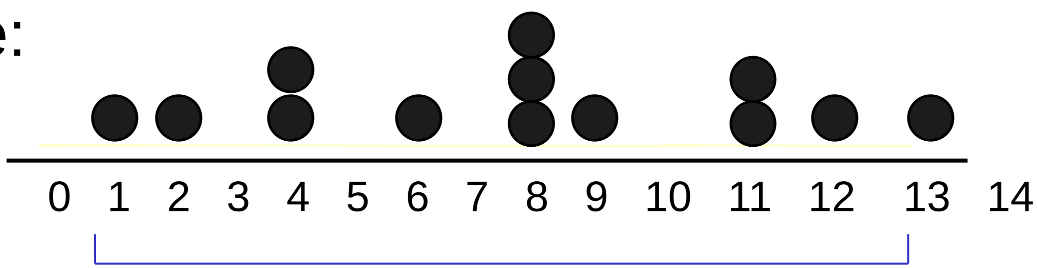
Measures of Variation: The Range

DCOVA

- Simplest measure of variation.
- Difference between the largest and the smallest values:

$$\text{Range} = x_{\text{max}} - x_{\text{min}}$$

Example:

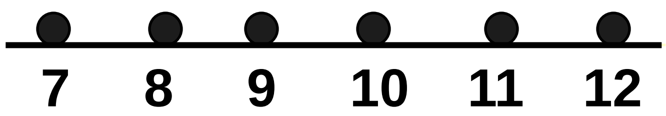


Range = 13 - 1 = 12

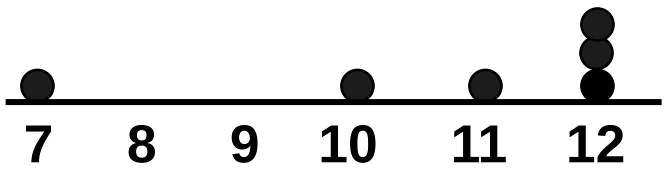
Measures of Variation:
Why The Range Can Be Misleading

DCOVA

- Does not account for how the data are distributed.



Range = 12 - 7 = 5



Range = 12 - 7 = 5

- Sensitive to outliers

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,**5**

Range = 5 - 1 = 4

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,**120**

Range = 120 - 1 = 119

Measures of Variation: The Sample Variance

DCOVA

- Average (approximately) of squared deviations of values from the mean.

  - Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

where $\bar{x}$ = arithmetic mean

n = sample size

x_i = i[th] value of the variable X

Calculation Formula of the Sample Variance

- The sample variance formula:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

can also, in order to facilitate computation, can also be expressed as follows.

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left[\sum_{i=1}^{n} x_i\right]^2\right)$$

Measures of Variation: The Sample Standard Deviation

- Most commonly used measure of variation.
- Shows variation about the mean.
- Is the square root of the variance.
- Has the same units as the original data.
- Sample Standard Deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left[\sum_{i=1}^{n} x_i\right]^2\right)}$$

Measures of Variation: The Sample Standard Deviation

Steps for Computing Standard Deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

1. Compute the difference between each value and the mean.
2. Square each difference.
3. Add the squared differences.
4. Divide this total by n-1 to get the sample variance.
5. Take the square root of the sample variance to get the sample standard deviation.

Measures of Variation: The Sample Standard Deviation

Steps for Computing Standard Deviation:

$$s = \sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left[\sum_{i=1}^{n} x_i\right]^2\right)}$$

1. Compute the square of each value.
2. Sum the computed squares from (1.).
3. Compute the square of the sum of all values.
4. Divide the answer from (3.) by the count (n) of all values.
5. Subtract answer (4.) from answer (2.) .
6. Divide this difference by $(n-1)$ to get the sample variance.
7. Take the square root of the sample variance to get the sample standard deviation.

Measures of Variation: Sample Standard Deviation Calculation Example (Method 1)

DCOVA

Sample Data (X)	10	12	14	15	17	18	18	24

$n = 8$ $Mean\ (\bar{x}) = 16$

$$s = \sqrt{\frac{(10-\bar{x})^2+(12-\bar{x})^2+(14-\bar{x})^2+\cdots+(24-\bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{(10-16)^2+(12-16)^2+(14-16)^2+\cdots+(24-16)^2}{8-1}}$$

$$= \sqrt{\frac{130}{7}} \approx 4.31$$

A measure of the “average” scatter around the mean.

Measures of Variation: Sample Standard Deviation Calculation Example (Method 2)

DCOVA

									Σ
Sample Data (X)	10	12	14	15	17	18	18	24	**128**
Squared Values (X^2)	100	144	196	225	289	324	324	576	**2178**

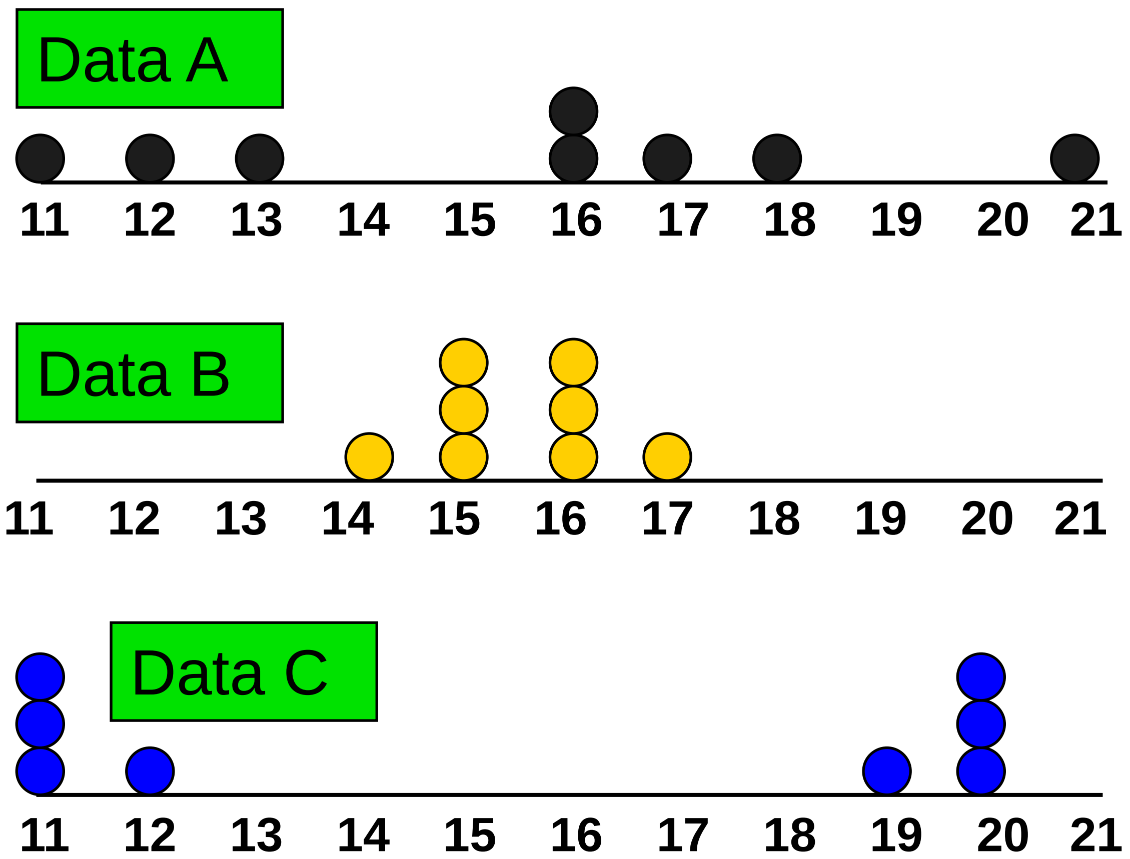
$$s = \sqrt{\frac{1}{8-1}\left(2178 - \frac{1}{8}(128)^2\right)} = \sqrt{\frac{1}{7}(2178 - 2048)}$$

$$= \sqrt{\frac{130}{7}} \approx 4.31$$

A measure of the "average" scatter around the mean.

Measures of Variation: Comparing Standard Deviations

DCOVA



Mean = 15.5
s = 3.338

Mean = 15.5
s = 0.926

Mean = 15.5
s = 4.567

Mean & Variance From Frequency Data

- Data are sometimes presented in terms of frequency of occurrence, for example, to ease data capturing.

- For example, let:

$\boldsymbol{X} = \{3, 1, 5, 3, \mathbf{2}, 4, \mathbf{2}, \mathbf{5}, 1, \mathbf{2}, \mathbf{2}, \mathbf{2}, \mathbf{5}, \mathbf{3}, \mathbf{2}\} \quad \equiv$

X	1	2	3	4	5
Frequency (F)	2	6	3	1	3

- In cases like this, calculating sample mean $(\bar{x})$ and sample variance (s^2) require a slightly different approach.

Mean & Variance From Frequency Data

						Σ
X	1	2	3	4	5	
Frequency (F)	2	6	3	1	3	**15**
FX	2	12	9	4	15	**42**
FX^2	2	24	27	16	75	**144**

$$n = \sum F = 2 + 6 + 3 + 1 + 3 = 15$$

$$\bar{x} = \frac{1}{n}\sum XF = \frac{1}{15}[(1 \times 2) + (2 \times 6) + (3 \times 3) + (4 \times 1) + (5 \times 3)] = 2.8$$

$$s^2 = \frac{1}{n-1}\sum F(X - \bar{x})^2 = \frac{1}{14}[2(1 - 2.8)^2 + 6(2 - 2.8)^2 + \cdots] \approx 1.9$$

$$s^2 = \frac{1}{n-1}\left[\sum f x^2 - \frac{1}{n}\left(\sum f x\right)^2\right] = \frac{1}{14}\left[144 - \frac{1}{15}(42)^2\right] \approx 1.9$$

Grouped Data Summary

- Other times data are presented in grouped format.

- For example, consider the data below of the ages of students in a first-year statistics class.

Age Group	16 - 18	19 - 21	22 - 24	25 - 27
Frequency (F)	12	18	8	2

- Estimate the mean and the variance of the student ages.

Grouped Data Summary

Age Group	Frequency (f)	Midpoint (x)	fx	x^2	fx^2
16 - 18	12	17	204	289	3468
19 - 21	18	20	360	400	7200
22 - 24	8	23	184	529	4232
25 - 27	2	26	52	676	1352
Total	$n = \mathbf{40}$		**800**		**16252**

$$\text{Mean } (\bar{x}) \approx \frac{\sum fx}{n} = \frac{800}{40} = 20 \text{ years}$$

$$\text{Variance } (s^2) \approx \frac{1}{n-1}\sum f(x-\bar{x})^2 = \frac{1}{n-1}\left[\sum fx^2 - \frac{1}{n}\left(\sum fx\right)^2\right]$$

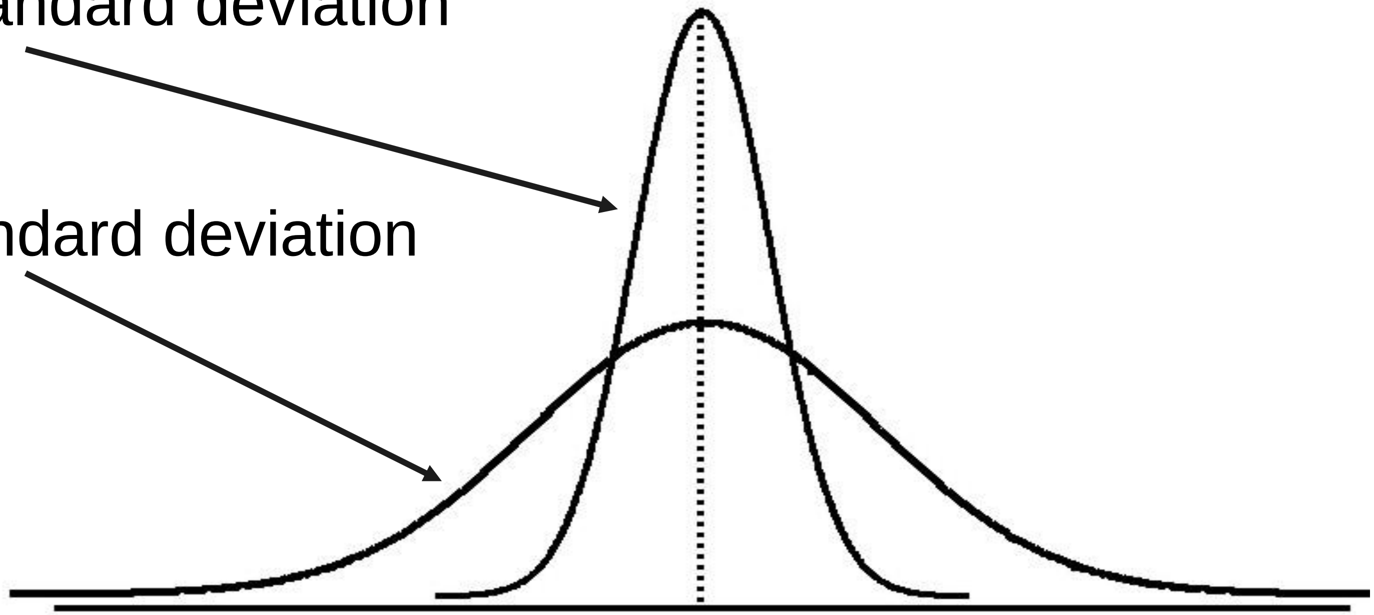
$$= \frac{1}{39}\left[16252 - \frac{1}{40}(800)^2\right] \approx 6.46$$

Measures of Variation: Comparing Standard Deviations

DCOVA

Smaller standard deviation

Larger standard deviation



Measures of Variation: Summary Characteristics

DCOVA

- The more the data are spread out, the greater the range, variance, and standard deviation.
- The more the data are concentrated, the smaller the range, variance, and standard deviation.
- If the values are all the same (no variation), all these measures will be zero.
- None of these measures are ever negative.

Measures of Variation: The Coefficient of Variation

DCOVA

- Measures relative variation.

- Always in percentage (%).

- Shows variation relative to mean.

- Can be used to compare the variability of two or more sets of data measured in different units.

$$CV = \left[\frac{s}{\bar{x}} \right] \times 100\%$$

Measures of Variation: Comparing Coefficients of Variation

DCOVA

- Stock A:
  - Mean price last year = R50.
  - Standard deviation = R5.

$$CV_A = \left(\frac{s}{\bar{x}}\right) \times 100\% = \frac{\text{R5}}{\text{R50}} \cdot 100\% = 10\%$$

- Stock B:
  - Mean price last year = R100.
  - Standard deviation = R5.

$$CV_B = \left(\frac{s}{\bar{x}}\right) \times 100\% = \frac{\text{R5}}{\text{R100}} \cdot 100\% = 5\%$$

Both stocks have the same standard deviation, but stock B is less variable relative to its mean price.

Measures of Variation: Comparing Coefficients of Variation (cont'd)

DCOVA

- Stock A:
  - Mean price last year = R50.
  - Standard deviation = R5.

$$CV_A = \left(\frac{s}{\bar{x}}\right) \times 100\% = \frac{\text{R5}}{\text{R50}} \times 100\% = 10\%$$

- Stock C:
  - Mean price last year = R8.
  - Standard deviation = R2.

$$CV_c = \left(\frac{s}{\bar{x}}\right) \times 100\% = \frac{\text{R2}}{\text{R8}} \times 100\% = 25\%$$

Stock C has a much smaller standard deviation but a much higher coefficient of variation

Locating Extreme Outliers: Z-Score

DCOVA

- To compute the Z**-score** of a data value, subtract the mean and divide by the standard deviation.
- The Z-score is the number of standard deviations a data value is from the mean.
- A data value is considered an extreme outlier if its Z-score is less than -3.0 or greater than +3.0.
- The larger the absolute value of the Z-score, the farther the data value is from the mean.

Locating Extreme Outliers: Z-Score

DCOVA

$$z = \frac{x - \bar{x}}{s}$$

where x represents the data value

$\bar{x}$ is the sample mean

s is the sample standard deviation

Locating Extreme Outliers: Z-Score

DCOVA

- Suppose the mean math SAT score is 490, with a standard deviation of 100.

- Compute the Z-score for a test score of 620.

$$z = \frac{x - \bar{x}}{s} = \frac{620 - 490}{100} = \frac{130}{100} = 1.30$$

A score of 620 is 1.30 standard deviations above the mean and would not be considered an outlier.

Shape of a Distribution

DCOVA

- Describes how data are distributed.

- Two useful shape related statistics are:
  - Skewness:
    - Measures the extent to which data values are not symmetrical.

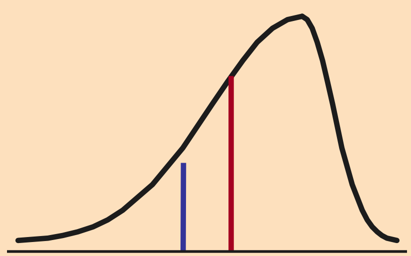
  - Kurtosis:
    - Kurtosis measures the “peakness” of the curve of the distribution—that is, how sharply the curve rises approaching the center of the distribution.

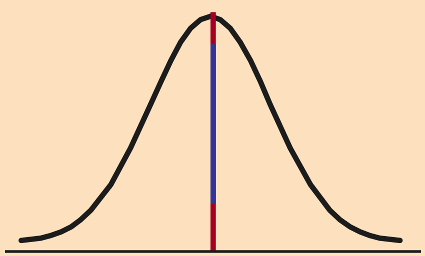
Shape of a Distribution (Skewness)

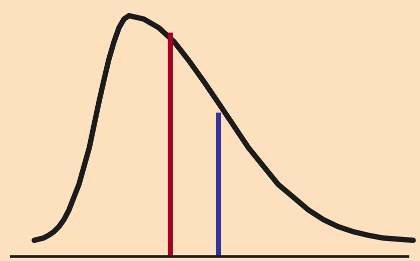
DCOVA

- Measures the extent to which data is not symmetrical.

Left-Skewed	Symmetric	Right-Skewed
Mean < Median	Mean = Median	Median < Mean



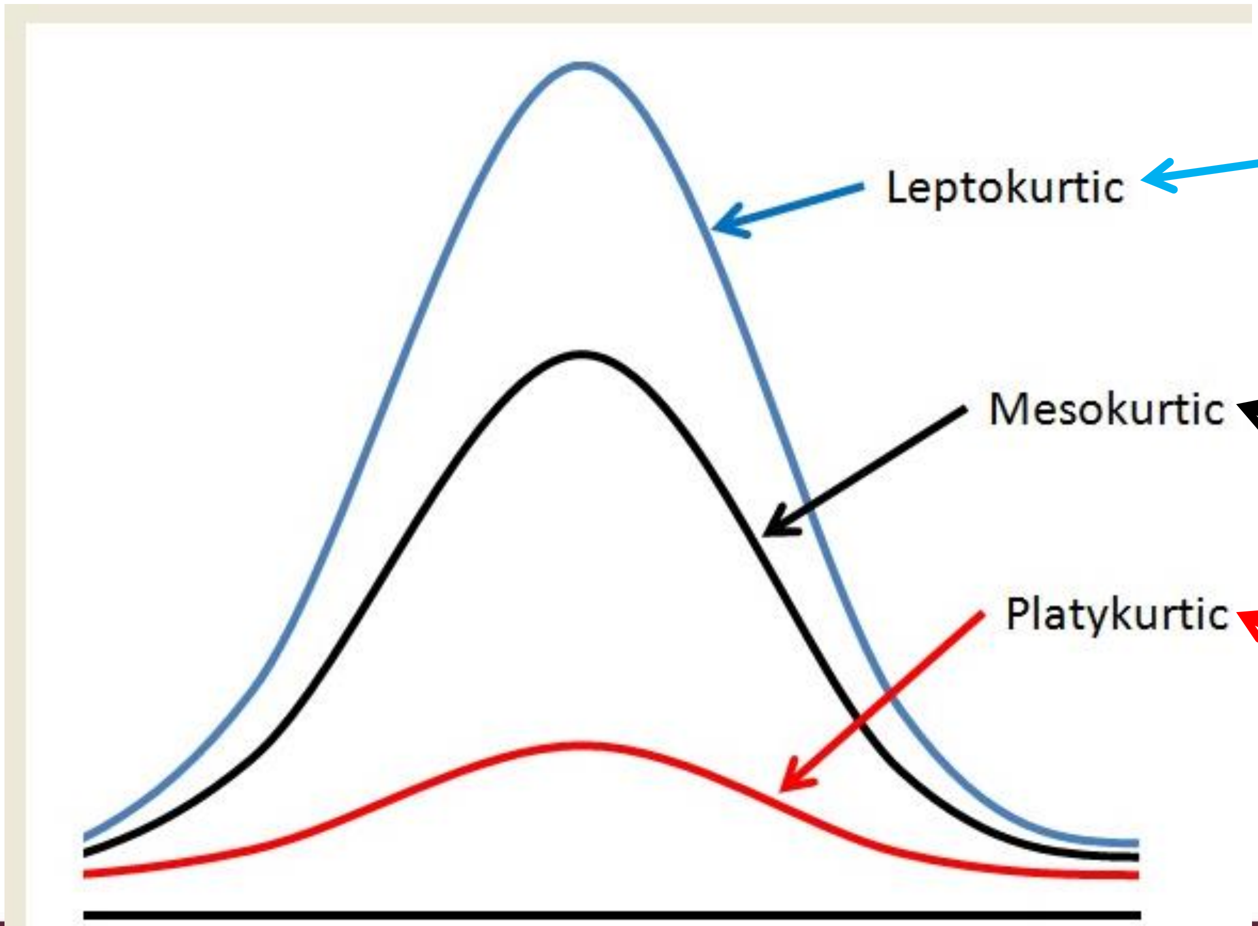




Skewness Statistic	< 0	0	>0

Shape of a Distribution -- Kurtosis measures how sharply the curve rises approaching the center of the distribution

DCOVA



Sharper Peak Than Bell-Shaped (Kurtosis > 0)

Bell-Shaped (Kurtosis = 0)

Flatter Than Bell-Shaped (Kurtosis < 0)

Exploring Numerical Data Using Quartiles

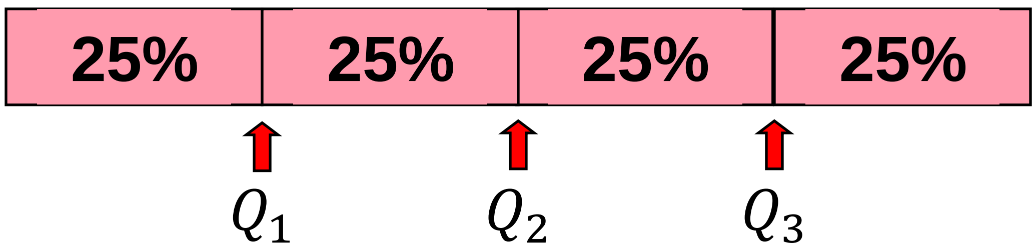
DCOVA

- Can visualise the distribution of the values for a numerical variable by computing:
  - The quartiles.
  - The five-number summary.
  - Constructing a boxplot.

Quartile Measures

DCOVA

- Quartiles split the ranked data into 4 segments with an equal number of values per segment.



  - The first quartile, Q_1, is the value for which 25% of the values are smaller and 75% are larger.
  - Q_2 is the same as the median (50% of the values are smaller and 50% are larger).
  - Only 25% of the values are greater than the third quartile.

Quartile Measures: Locating Quartiles

DCOVA

Find a quartile by determining the value in the appropriate position in the ranked data, where:

First quartile position: $Q_1 = (n+1)/4$ ranked value.

Second quartile position: $Q_2 = (n+1)/2$ ranked value.

Third quartile position: $Q_3 = 3(n+1)/4$ ranked value.

where **n** is the number of observed values.

Quartile Measures: Calculation Rules

DCOVA

- When calculating the ranked position use the following rules:
  - If the result is a whole number then it is the ranked position to use.
  - If the result is a fractional half (e.g. 2.5, 7.5, 8.5, etc.) then average the two corresponding data values.
  - If the result is not a whole number and not a fractional half then round the result to the nearest integer to find the ranked position.

Quartile Measures
Calculating The Quartiles: Example

DCOVA

Sample Data in Ordered Array: 11 12 13 16 16 17 18 21 22

(n = 9)

Q_1 is in the (9+1)/4 = 2.5th position of the ranked data,

so **Q_1 = (12+13)/2 = 12.5.**

Q_2 is in the (9+1)/2 = 5th position of the ranked data,

so **Q_2 = median = 16.**

Q_3 is in the 3(9+1)/4 = 7. 5th position of the ranked data,

so **Q_3 = (18+21)/2 = 19.5.**

Q_1 and Q_3 are measures of non-central location.
Q_2 = median, is a measure of central tendency.

Quartile Measures: The Interquartile Range (IQR)

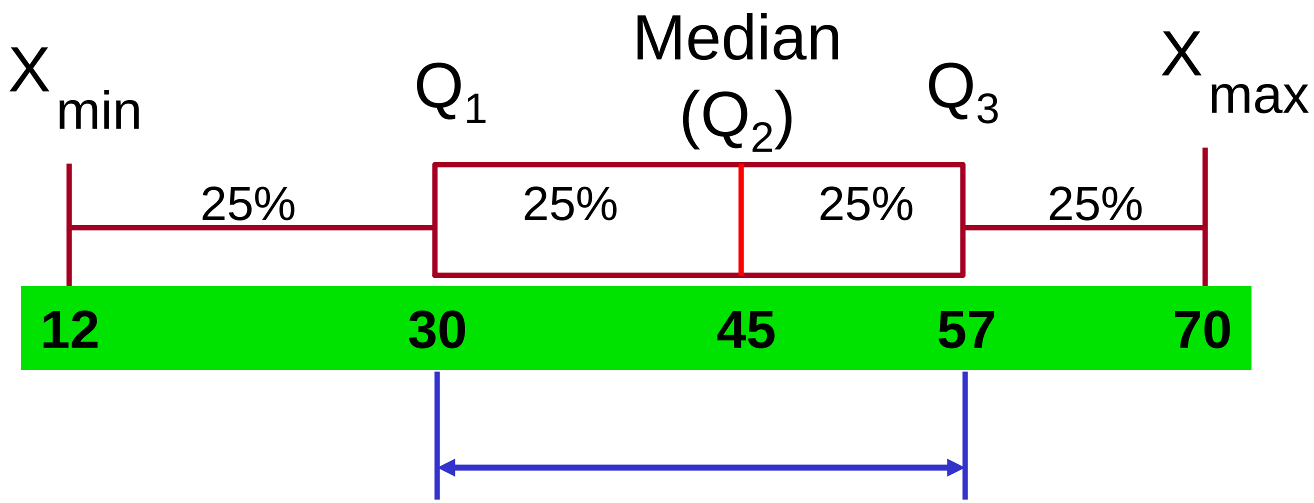
DCOVA

- The IQR is $Q_3 – Q_1$ and measures the spread in the middle 50% of the data.
- The IQR is also called the midspread because it covers the middle 50% of the data.
- The IQR is a measure of variability that is not influenced by outliers or extreme values.
- Measures like Q_1, Q_3, and IQR that are not influenced by outliers are called resistant measures.

Calculating The Interquartile Range

DCOVA

Example:



Interquartile range
= 57 – 30 = 27

The Five Number Summary

DCOVA

The five numbers that help describe the center, spread and shape of data are:

- x_{min}.
- First Quartile (Q_1).
- Median (Q_2).
- Third Quartile (Q_3).
- x_{max}.

Relationships among the five-number summary and distribution shape

DCOVA

Left-Skewed	Symmetric	Right-Skewed
Median – x_{min} $>$ x_{max} – Median	Median – x_{min} $\approx$ x_{max} – Median	Median – x_{min} $<$ x_{max} – Median
Q_1 – x_{min} $>$ x_{max} – Q_3	Q_1 – x_{min} $\approx$ x_{max} – Q_3	Q_1 – x_{min} $<$ x_{max} – Q_3
Median – Q_1 $>$ Q_3 – Median	Median – Q_1 $\approx$ Q_3 – Median	Median – Q_1 $<$ Q_3 – Median

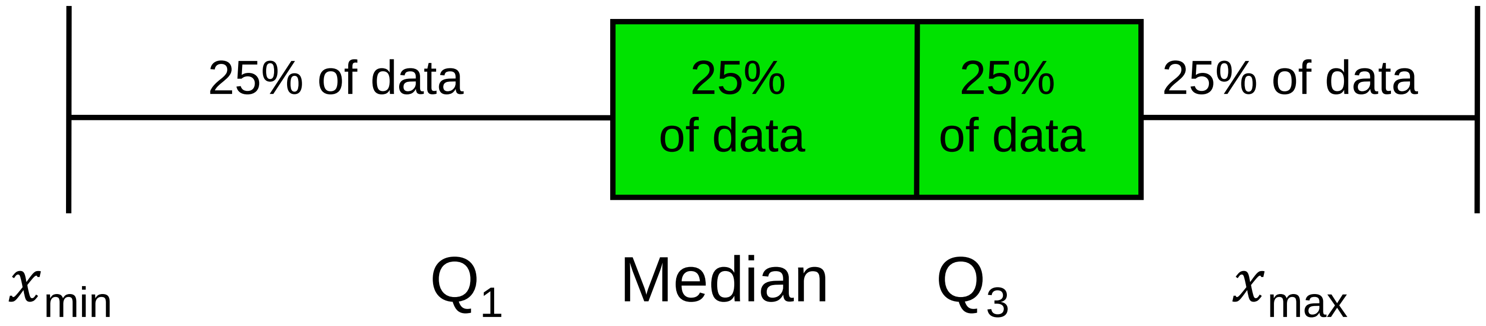
Five Number Summary and The Boxplot

DCOVA

- The Boxplot: A Graphical display of the data based on the five-number summary:

x_{min} -- Q_1 -- Median -- Q_3 -- x_{max}

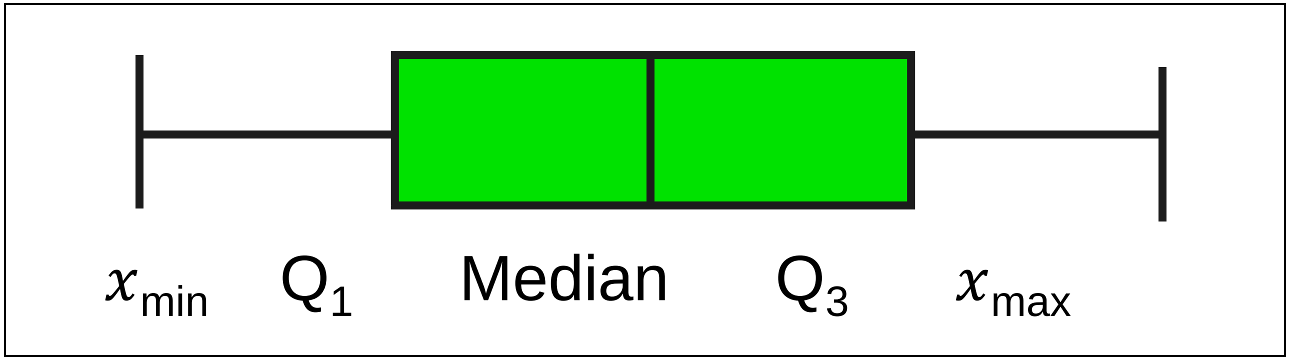
Example:



Five Number Summary: Shape of Boxplots

DCOVA

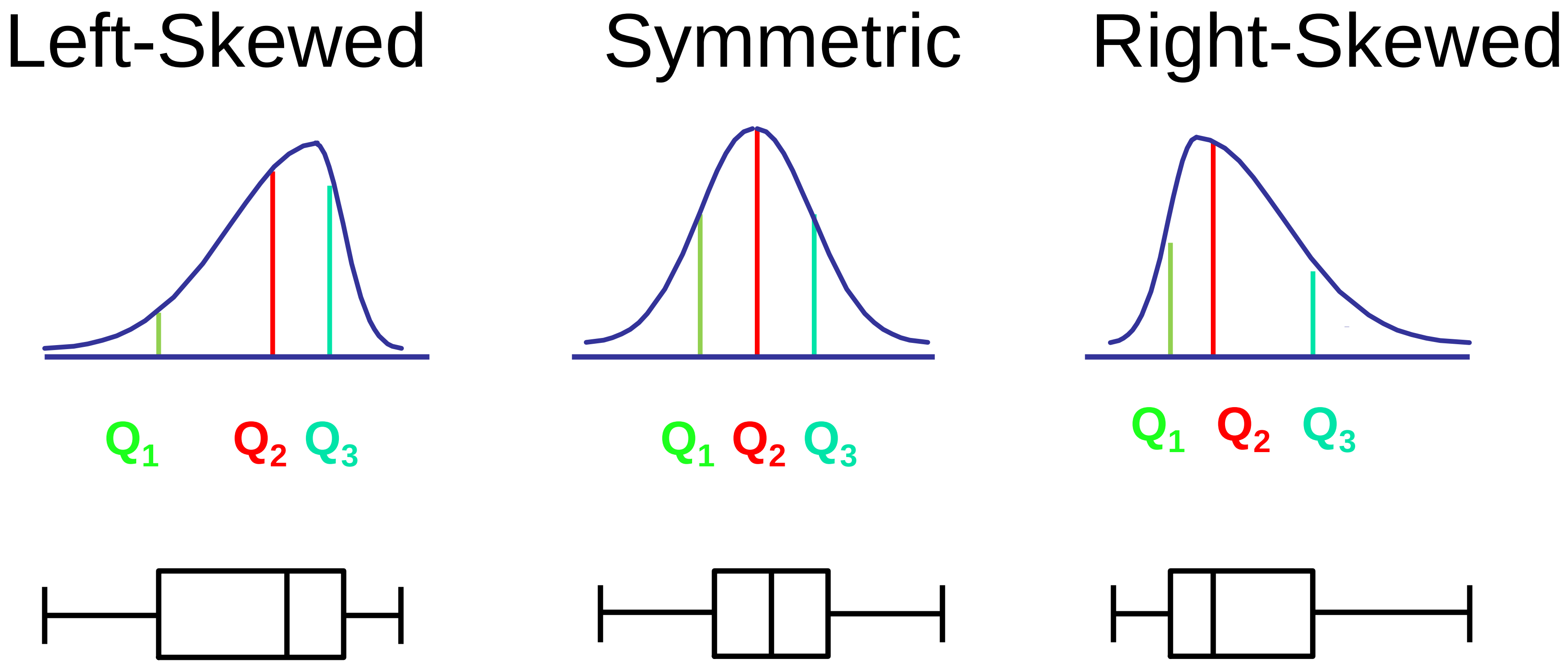
- If data are symmetric around the median then the box and central line are centered between the endpoints.



- A Boxplot can be shown in either a vertical or horizontal orientation.

Distribution Shape and The Boxplot

DCOVA



Boxplot Example

DCOVA

- Below is a Boxplot for the following data:

x_{min}		Q_1			Q_2 / Median			Q_3		x_{max}
0	2	2	2	3	3	4	5	5	9	27

- The data are right skewed, as the plot depicts.

Numerical Descriptive Measures for a Population

DCOV<u>A</u>

- Descriptive statistics discussed previously described a *sample*, not the *population*.

- Summary measures describing a population, called **parameters**, are denoted with Greek letters.

- Important population parameters are the population mean, variance, and standard deviation.

Numerical Descriptive Measures for a Population: The mean μ

DCOVA

- The population mean is the sum of the values in the population divided by the population size, N.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i = \frac{x_1 + x_2 + \cdots + x_N}{N}$$

where μ = population mean

N = population size

x_i = i^{th} value of the variable X

Numerical Descriptive Measures For A Population: The Variance σ^2

DCOVA

- Average of squared deviations of values from the mean.
  - Population variance:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 = \frac{1}{N}\left(\sum_{i=1}^{N} x_i^2 - \frac{1}{N}\left(\sum_{i=1}^{N} x_i\right)^2\right)$$

where μ = population mean

N = population size

x_i = i^{th} value of the variable X

Numerical Descriptive Measures For A Population: The Standard Deviation σ

DCOVA

- Most commonly used measure of variation.
- Shows variation about the mean.
- Is the square root of the population variance.
- Has the same units as the original data.
  - Population standard deviation:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2} = \sqrt{\frac{1}{N}\left(\sum_{i=1}^{N}x_i^2 - \frac{1}{N}\left(\sum_{i=1}^{N}x_i\right)^2\right)}$$

Sample statistics versus population parameters

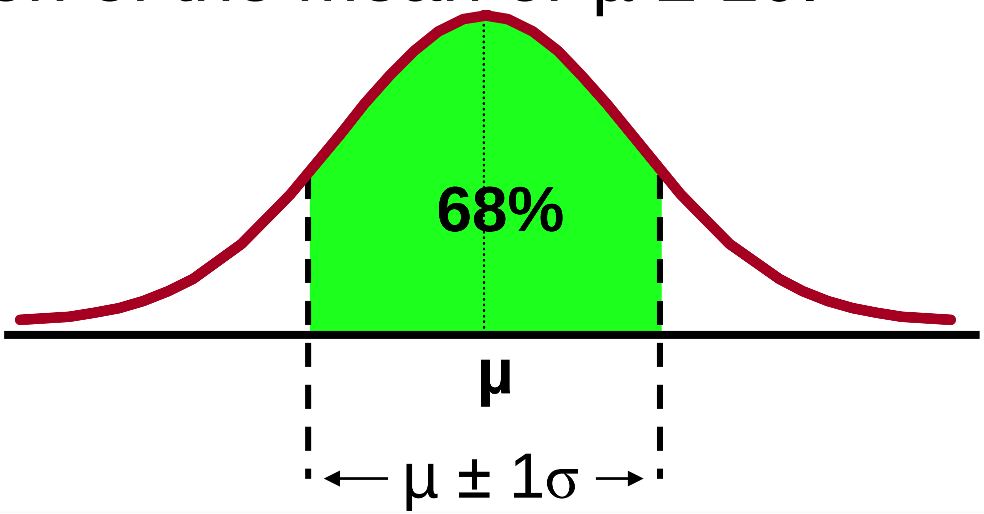
DCOVA

Measure	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s

The Empirical Rule

DCOVA

- The empirical rule approximates the variation of data in a symmetric mound-shaped distribution.
- Approximately 68% of the data in a symmetric mound shaped distribution is within 1 standard deviation of the mean or $\mu \pm 1\sigma$.

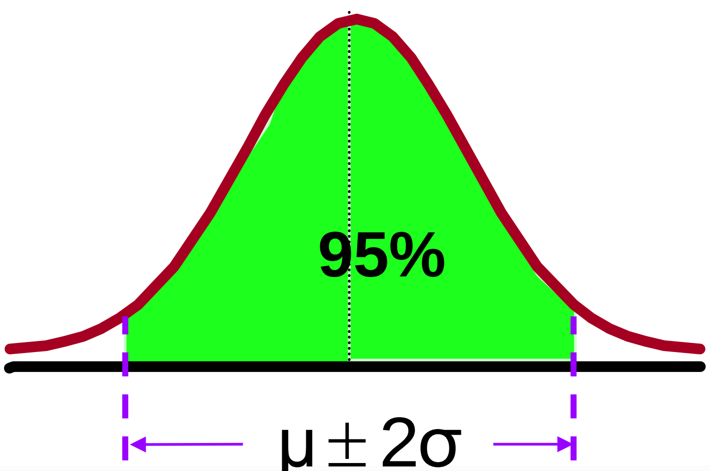


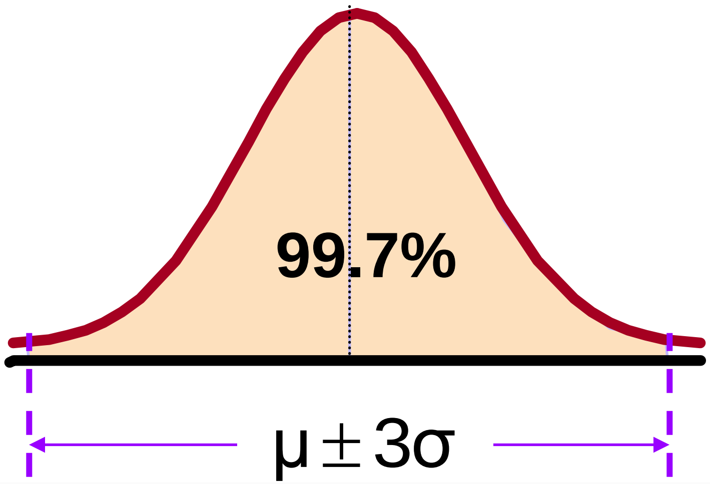
The Empirical Rule

DCOVA

- Approximately 95% of the data in a symmetric mound-shaped distribution lies within two standard deviations of the mean, or $\mu \pm 2\sigma$.

- Approximately 99.7% of the data in a symmetric mound-shaped distribution lies within three standard deviations of the mean, or $\mu \pm 3\sigma$.





Using the Empirical Rule

DCOVA

- Suppose that the variable Math SAT scores is bell-shaped with a mean of 500 and a standard deviation of 90. Then:

  - Approximately 68% of all test takers scored between 410 and 590, (500 ± 90).

  - Approximately 95% of all test takers scored between 320 and 680, (500 ± 180).

  - Approximately 99.7% of all test takers scored between 230 and 770, (500 ± 270).

Chebyshev's Rule

DCOVA

- Regardless of how the data are distributed, at least $(1 - 1/k^2) \times 100\%$ of the values will fall within k standard deviations of the mean (for $k > 1$).

  - Examples:

	At least	Within
$(1 - 1/2^2) \times 100\% =$	75%	$k=2$ $(\mu \pm 2\sigma)$
$(1 - 1/3^2) \times 100\% =$	88.89%	$k=3$ $(\mu \pm 3\sigma)$

We Discuss Two Measures Of The Relationship Between Two Numerical Variables

- Scatter plots allow you to visually examine the relationship between two numerical variables and now we will discuss two quantitative measures of such relationships.

- The Covariance.
- The Coefficient of Correlation.

The Covariance Measures Strength Of The Linear Relationship Between Two Numerical Variables (X & Y)

- The sample covariance: DCOVA

$$\mathrm{cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- Only concerned with the strength of the relationship.

- No causal effect is implied.

Calculation Formula of the Covariance

- In order to facilitate computation, the covariance formula:

$$\operatorname{cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

can also be expressed as follows.

$$\operatorname{cov}(X,Y) = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i y_i - \frac{1}{n}\left[\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i\right]\right)$$

Interpreting Covariance

DCOVA

- **Covariance** between two variables:

$\text{cov}(X,Y) > 0 \longrightarrow$ X and Y tend to move in the same direction.

$\text{cov}(X,Y) < 0 \longrightarrow$ X and Y tend to move in opposite directions.

$\text{cov}(X,Y) = 0 \longrightarrow$ X and Y are independent.

- The covariance has a major flaw:
  - It is not possible to determine the relative strength of the relationship from the size of the covariance.

Measuring The Relative Strength Relationships: Coefficient of Correlation

DCOVA

- Measures the relative strength of the linear relationship between two numerical variables.
- Sample coefficient of correlation:

$$r = \frac{cov(\mathrm{X,Y})}{s_x s_y}$$

where,

$$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$s_y = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

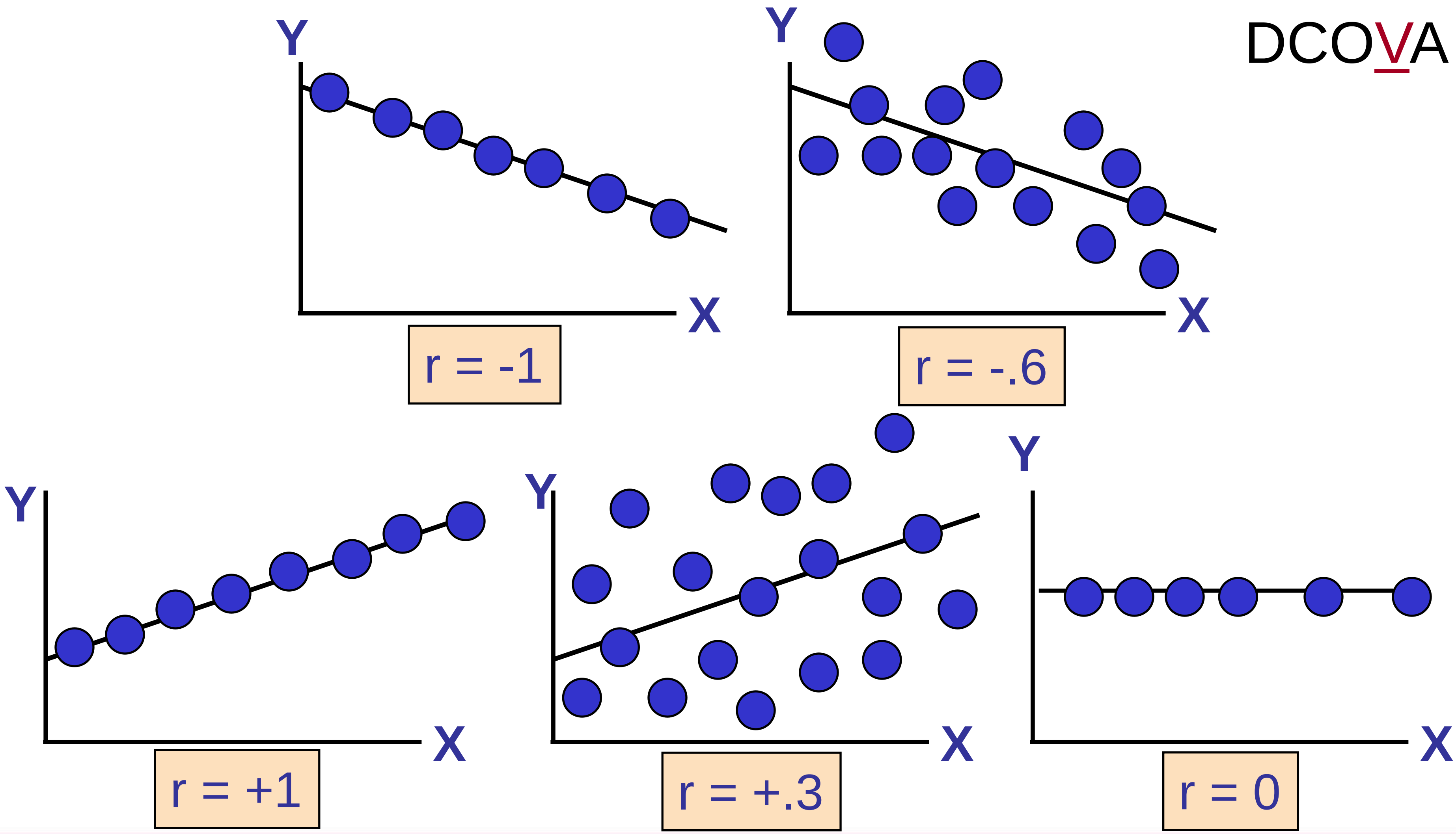
Features of the Coefficient of Correlation

DCOVA

- The population coefficient of correlation is referred as ρ.
- The sample coefficient of correlation is referred to as r.
- Either ρ or r have the following features:
  - Unit free.
  - Range between –1 and 1.
  - The closer to –1, the stronger the negative linear relationship.
  - The closer to 1, the stronger the positive linear relationship.
  - The closer to 0, the weaker the linear relationship.

Scatter Plots of Sample Data with Various Coefficients of Correlation

DCOVA



Pitfalls in Numerical Descriptive Measures

DCOVA

- Data analysis is objective:
  - Should report the summary measures that best describe and communicate the important aspects of the data set.

- Data interpretation is subjective:
  - Should be done in fair, neutral and clear manner.

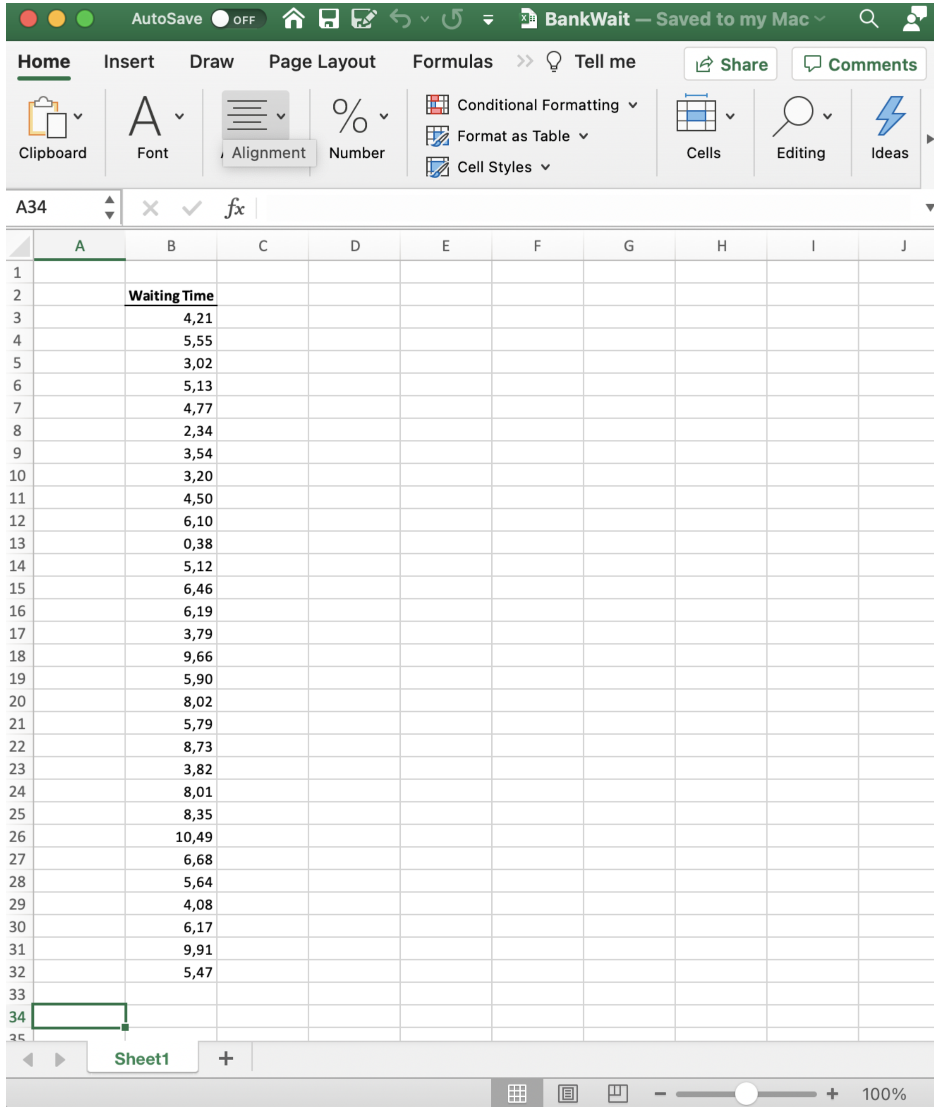
Ethical Considerations

DCOVA

Numerical descriptive measures:

- Should document both good and bad results.
- Should be presented in a fair, objective and neutral manner.
- Should not use inappropriate summary measures to distort facts.

Excel: Bank Waiting Times Data



	Waiting Time
3	4,21
4	5,55
5	3,02
6	5,13
7	4,77
8	2,34
9	3,54
10	3,20
11	4,50
12	6,10
13	0,38
14	5,12
15	6,46
16	6,19
17	3,79
18	9,66
19	5,90
20	8,02
21	5,79
22	8,73
23	3,82
24	8,01
25	8,35
26	10,49
27	6,68
28	5,64
29	4,08
30	6,17
31	9,91
32	5,47

Excel: Bank Waiting Times Data

Let $R_{start}:R_{end}$ denote a selection of the cells that contain the data to be analysed.

- Summation = `SUM(`$R_{start}:R_{end}$`)`.
- Mean = `AVERAGE(`$\mathrm{R_{start}:R_{end}}$`)`.
- Median = `MEDIAN(`$\mathrm{R_{start}:R_{end}}$`)`.
- Data Count (n) = `COUNT(`$\mathrm{R_{start}:R_{end}}$`)`

Excel: Bank Waiting Times Data

- Geometric Mean = GEOMEAN($\mathrm{R_{start}}$:$\mathrm{R_{end}}$).

- Variance = VAR(R_{start}:R_{end}).

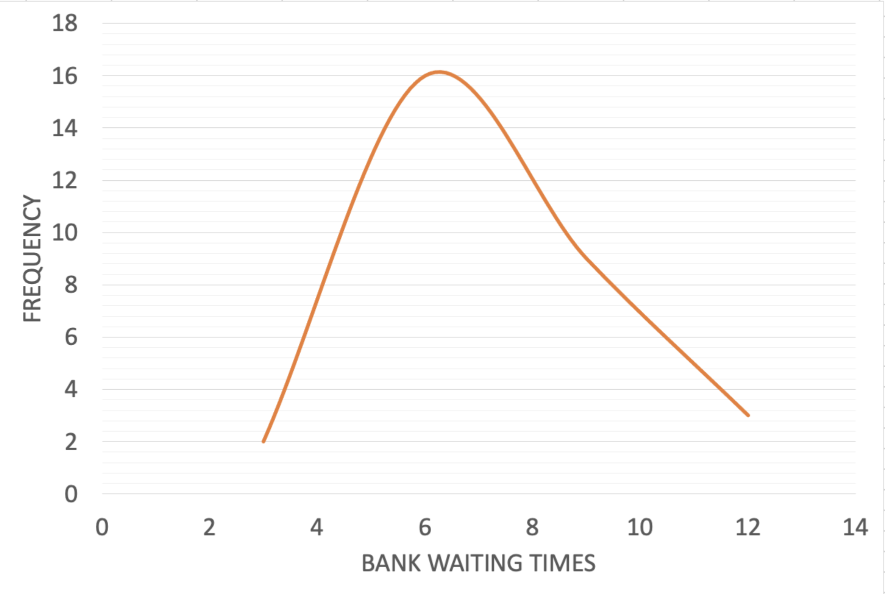
- Standard Deviation $[s(X)]$ = STDEV($\mathrm{R_{start}}$:$\mathrm{R_{end}}$).

- Z-Score = STANDARDIZE(R_{start}:R_{end}; $\bar{x}$; $s(X)$).

- Five-number Summary = QUARTILE(R_{start}:R_{end}; q)
  - Minimum: q=0
  - First quartile: q=1
  - Second quartile: q=2
  - Third quartile: q=3
  - Maximum: q=4

Excel: Bank Waiting Times Data

	B	D	F	H
2	Waiting Time			Z-Score:
3	4,21	Summation:	Minimum:	-0,64
4	5,55	171,02	0,38	-0,06
5	3,02			-1,15
6	5,13	Mean:	First Quartile:	-0,24
7	4,77	5,70	4,11	-0,40
8	2,34			-1,44
9	3,54	Median:	Second Quartile:	-0,92
10	3,20	5,60	5,79	-1,07
11	4,50			-0,51
12	6,10	Data Count (n):	Third Quartile:	0,17
13	0,38	30	8,01	-2,28
14	5,12			-0,25
15	6,46	Geometric Mean:	Maximum:	0,33
16	6,19	5,04	10,49	0,21
17	3,79			-0,82
18	9,66	Variance:	Skewness:	1,69
19	5,90	5,46	0	0,09
20	8,02			0,99
21	5,79	Standard Deviation:	Kurtosis:	0,04
22	8,73	2,34	0	1,30
23	3,82			-0,81
24	8,01			0,99
25	8,35			1,13
26	10,49			2,05
27	6,68			0,42
28	5,64			-0,03
29	4,08			-0,69
30	6,17			0,20
31	9,91			1,80
32	5,47			-0,10



Sheet1

Chapter Summary

In this chapter we have discussed:

- Describing the properties of central tendency, variation, and shape in numerical variables.
- Constructing and interpreting a boxplot.
- Computing descriptive summary measures for a population.
- Calculating the covariance and the coefficient of correlation.